



V-Guard Sentinel

The inverter already knows when the power goes. Sentinel teaches it what to keep on — and when its battery will quit.

Track 4 · From Smart Products to Intelligent Products

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Finale · Kochi · 24 September 2026



10:03 pm. The battery dies — with no warning.

Three hours into a power cut, Home 6 in our village goes dark. Home 1 has the same battery and appliances — and stays lit.



85 %

of Indian households face a power cut every day (LocalCircles, 2023)

×2 wear

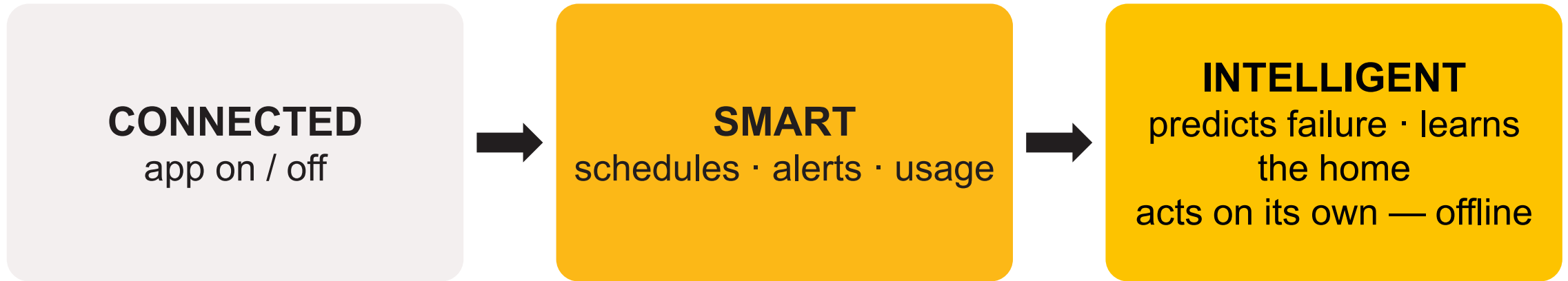
every +10 °C: lead-acid life halves with heat — the Indian summer is the battery's enemy

≈ 1.5 %

of V-Guard revenue goes to warranty (≈ ₹69 cr, FY24)

Sentinel turns a smart inverter into an intelligent one.

Smart products take commands. Intelligent products predict, decide and act — without the cloud.

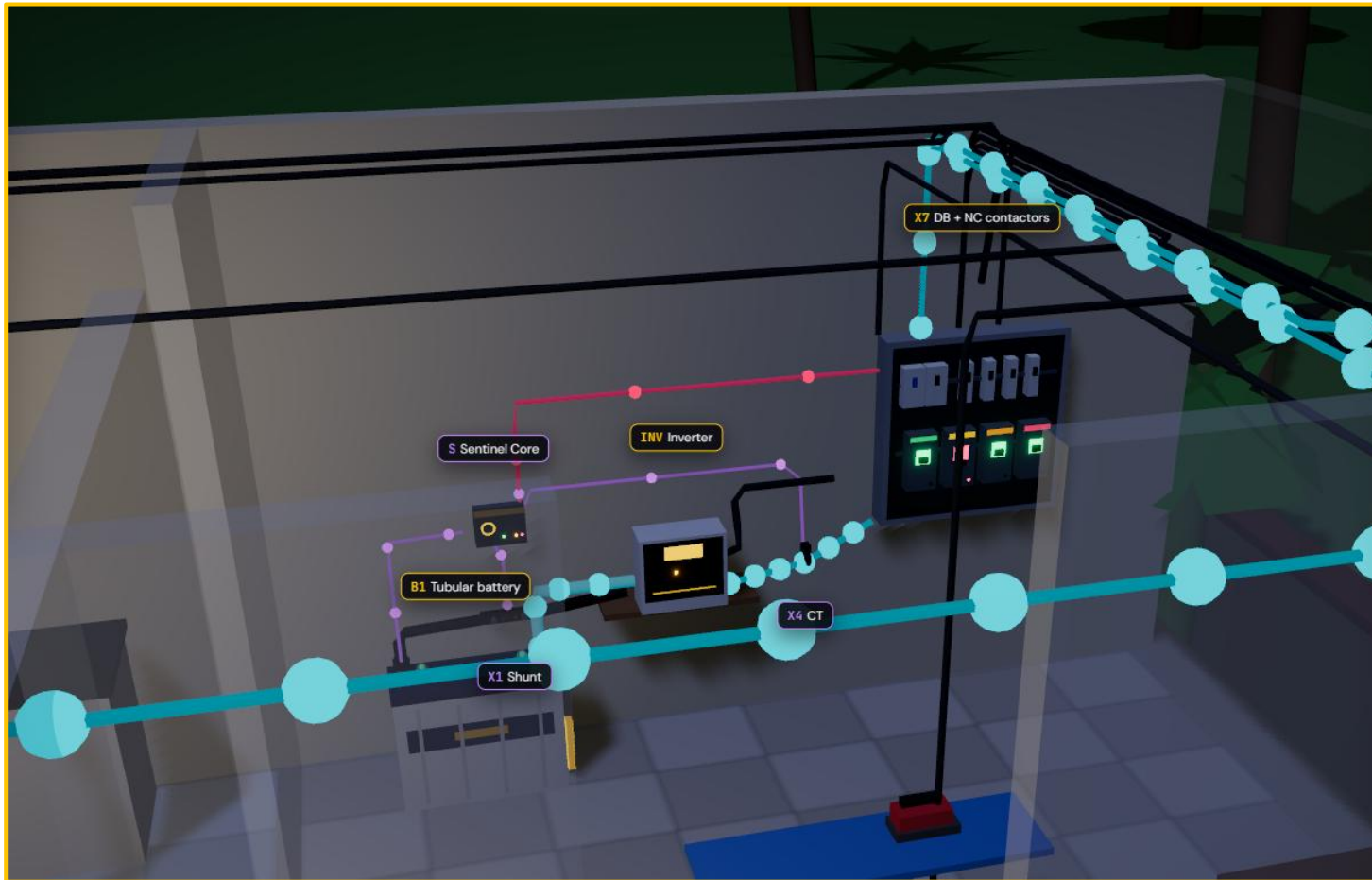


← Sentinel

- 1 Embedded** a daughter-board inside new V-Guard inverters
- 2 Retrofit** a 30-minute electrician fit on the installed base
- 3 Offline-first** every core function works with zero internet

One small box reads the battery and switches the loads.

Nothing on the battery or the inverter is modified on the retrofit — only the negative cable passes through a shunt.



- 1 Shunt in the battery – lead**
battery current → INA228
- 2 NTC on the battery post**
temperature: the #1 ageing driver
- 3 CT + voltage tap on AC-OUT**
real & reactive power → ATM90E32AS
- 4 4 NC contactors in the DB**
coil off = load ON, always
- 5 ESP32-S3 Sentinel Core**
90 × 70 × 35 mm, powered by the battery

Four jobs, one board — all on the ESP32, no cloud needed.

1

Battery health

Predicts a replacement window — 'replace in 6–14 weeks'

2

Habit autopilot

Keeps the fridge, lights and router alive during a cut

3

Energy coach + grid shield

Names the big appliances; logs every sag and swell

4

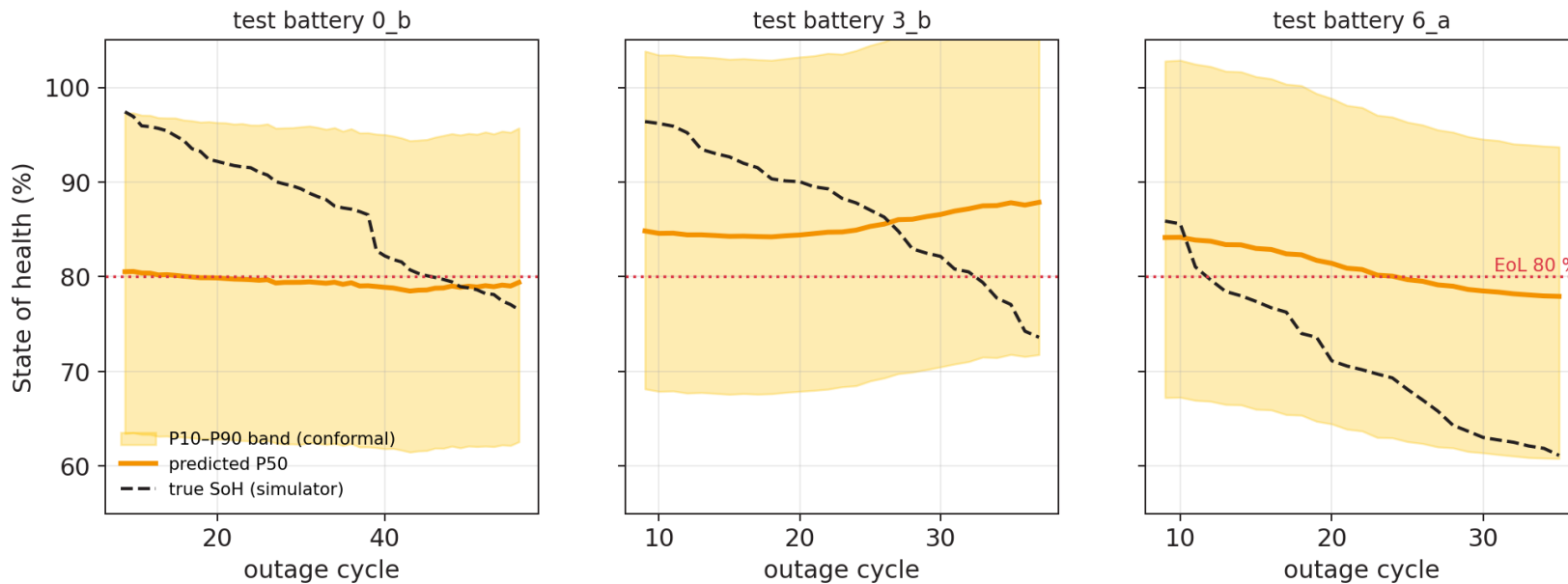
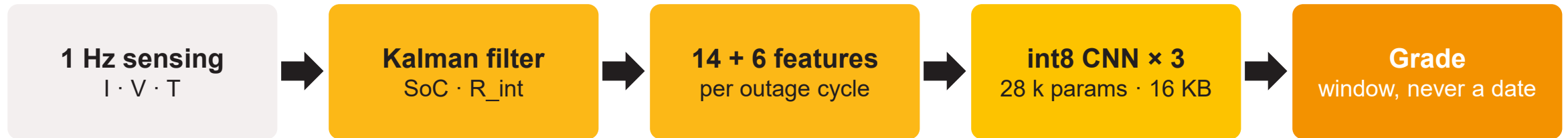
Signed health log

Tamper-proof record — warranty decided by evidence

Same core, same sensors, different models → the platform extends to pumps, stabilisers and water heaters.

It gives a window, not a date: "replace in 6–14 weeks".

Physics first, learning second — every step runs on the ESP32.



8.2 pt

SoH error on 3 never-seen test batteries (synthetic data — honest, not flattering)

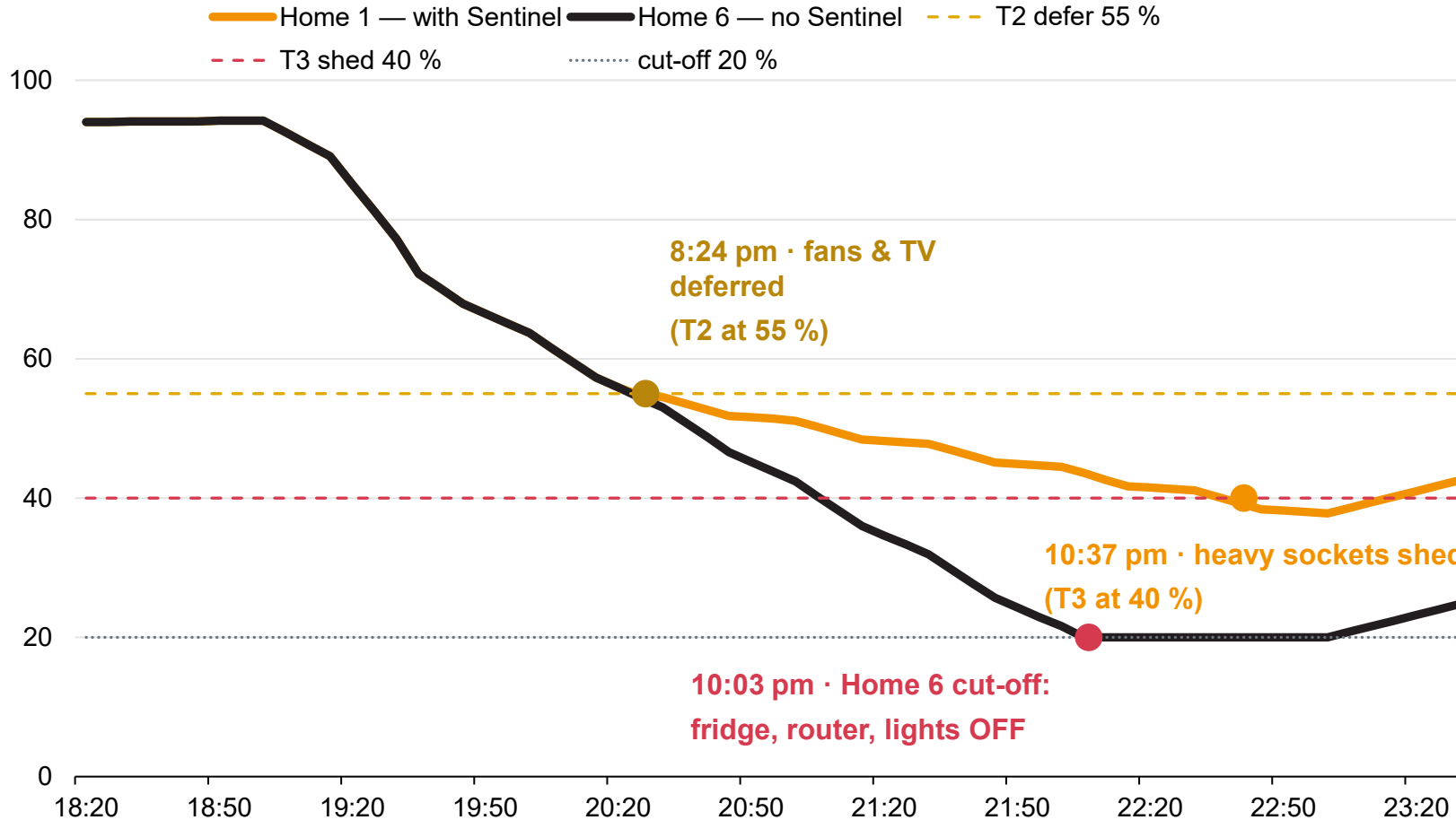
+0.01 pt

accuracy lost going float → int8 — it fits on the chip for free

Band is calibrated but wide today; the Kochi aging campaign narrows it.

Same outage, same battery: only Home 1 keeps its fridge on.

Battery charge through the 7–11 pm cut, straight from our simulation engine, using our specified shed thresholds.



57 min

Home 6 sat in the dark until mains returned at 11 pm

0 min

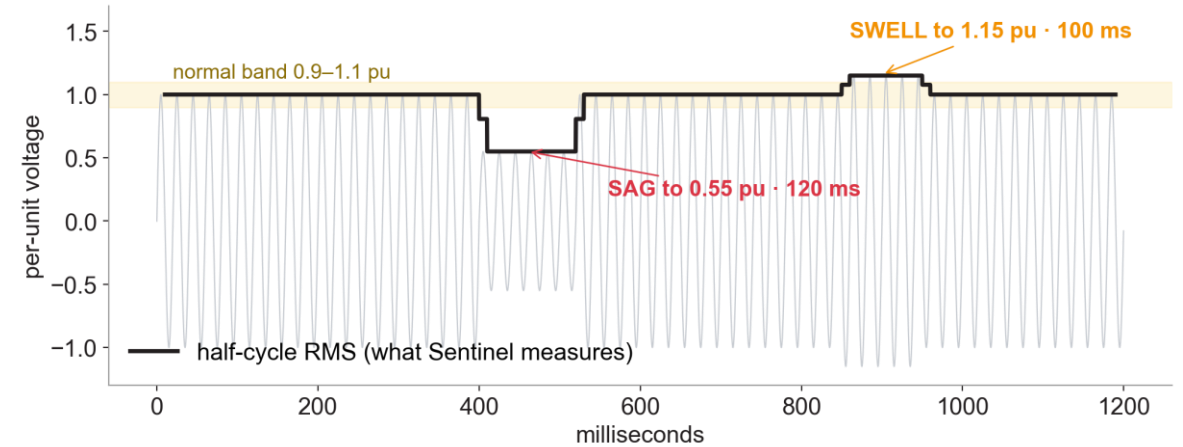
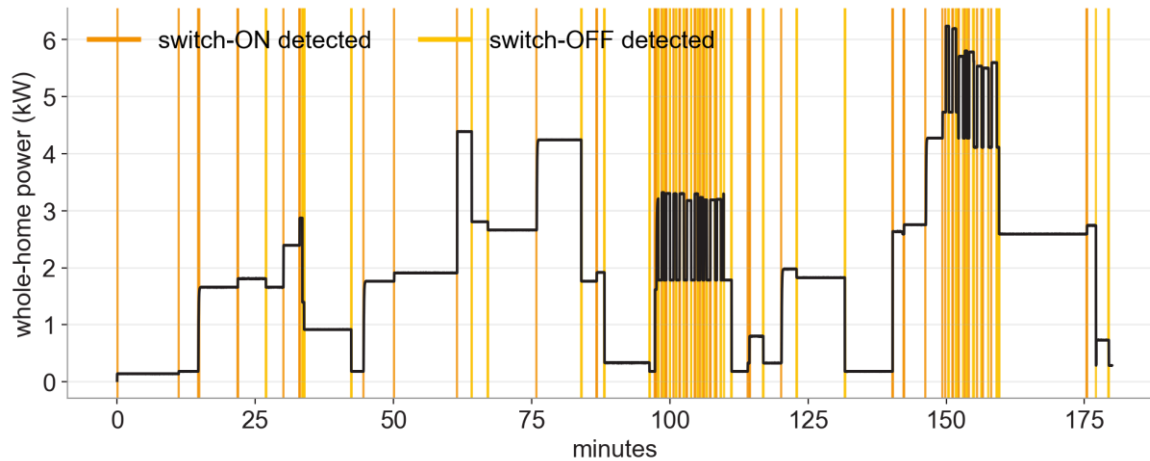
Home 1's fridge, router and hall light were never off

Coil off = load on.

NC contactors, a hardware supervisory timer and a medical jumper: any crash or reset brings every load back.

One sensor saves energy and catches a misbehaving grid.

Both run on the metering front-end (ATM90E32AS) and the isolated voltage tap (AMC1311) — outputs of the team's own detector code.

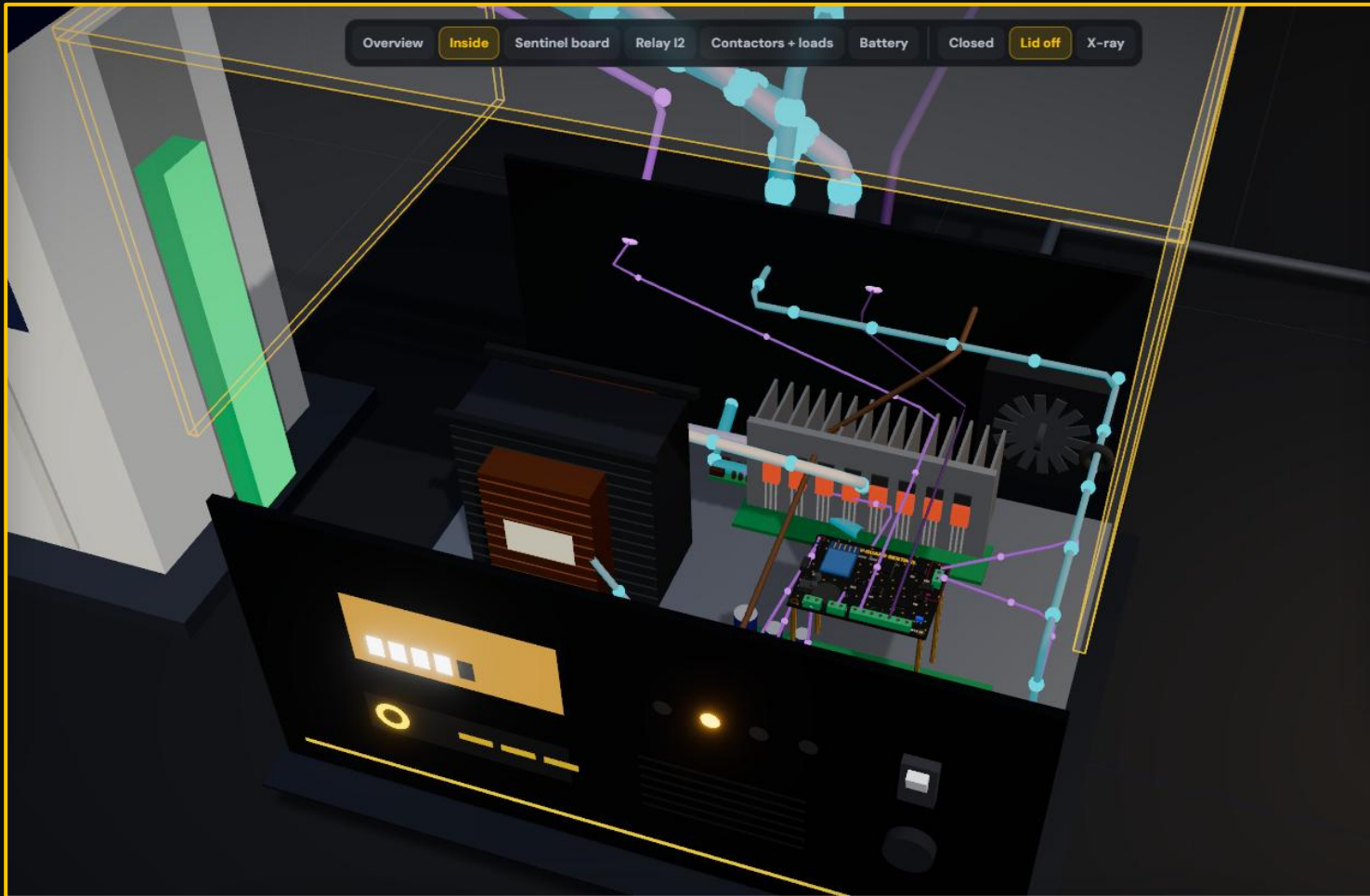


Energy Coach 123 switch-ON/OFF events found in 3 h of whole-home power; each becomes a 13-number signature. Fridge F1 0.97, iron 1.00.

Grid Shield a 120 ms sag to 0.55 pu measured as 0.54; a 100 ms swell to 1.15 pu as 1.16. 13 / 13 dips caught. Logged, dated, signed.

Inside the inverter, Sentinel taps what's already there.

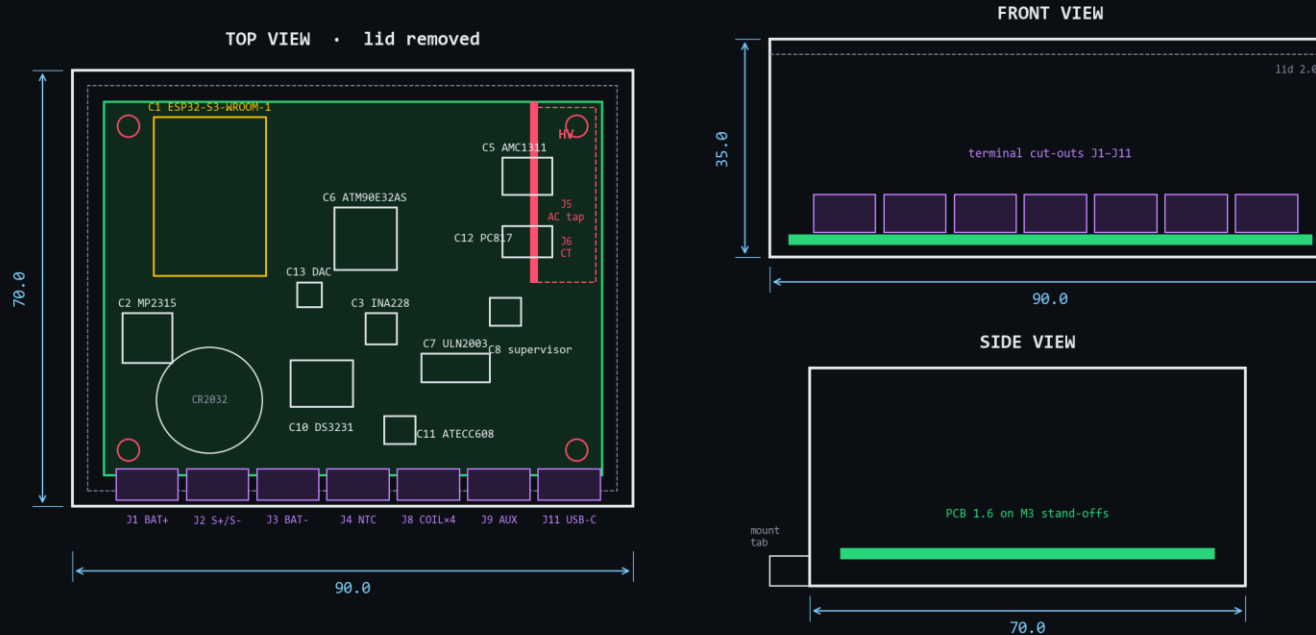
Our 3D proof of concept of the Embedded SKU: lid off, every part clickable, live power flow.



- 1 Inverter's own shunt (I8)**
→ INA228 · battery current
- 2 Mode / mains line**
→ PC817 opto · outage vote s2
- 3 SG3525 charge feedback**
→ MCP4725 DAC · temperature-compensated charging
- 4 AC-OUT rail**
→ AMC1311 + ATM90E32AS · V, P, Q
- 5 4 coil outputs**
→ NC contactor panel · T1 / T2 / T3 / medical

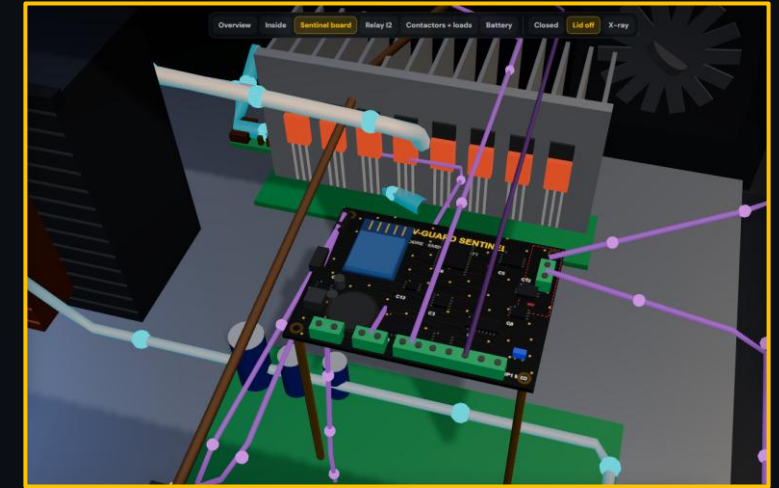
Designed to be built: 90 × 70 × 35 mm, HV zone fenced off.

Mechanical drawing from our final spec, and the same board as a 3D render in the simulation.



V-GUARD SENTINEL CORE – ENCLOSURE & PCB

Enclosure 90 × 70 × 35 mm · PC/ABS UL94 V-0 · wall 2.5 · IP20
PCB 80 × 60 × 1.6 · 4× M3 (Ø3.5) at 4 mm inset
Battery current never enters the PCB – only shunt mV (J2)
AMC1311 + PC817 straddle the HV isolation moat
Coil off = load on: ULN2003 + supervisory timer + JP1
Units mm · per prototype/02-03 (round 3) · Rev T1



Battery current never enters the PCB
only the shunt's millivolts do.

Isolation moat

AMC1311 and PC817 straddle the HV zone.

UL94 V-0 PC/ABS, 4× M3

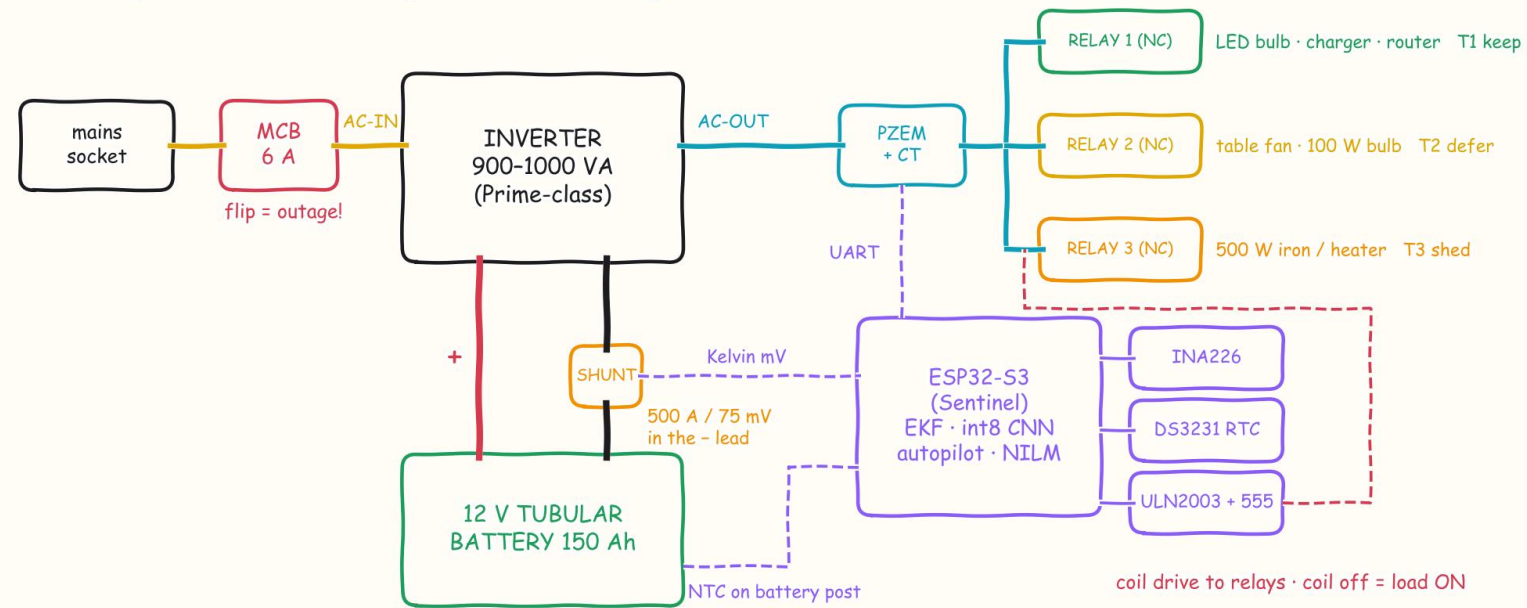
DIN or screw mount beside the inverter.

And this is exactly how the bench is wired.

Real battery, real inverter, off-the-shelf parts. Every component is available from Indian distributors today.

Tier-0 bench — how we wire it

real battery · real inverter · three relay-switched tiers · nothing on the internet



Reset the ESP32 while T3 is shed → relay 3 re-closes in about a second.

Let's watch it happen — live.

Sentinel Live runs offline in one HTML file. Three moments to watch for:



1 · The village

Feeder cuts at 7 pm. Watch Home 6 go dark at 10:03 while its twin stays lit.



2 · Flip the MCB

Relay swings in slow motion: 8 ms gap on the scope, outage confirmed in 1.5 s.



3 · Crash it

Drop SoC below 40 % → the iron sheds. Reset the chip → every load comes back.

What's proven, what's synthetic, and what still needs hardware.

We'd rather you trust three honest columns than one shiny number.

245

automated tests, all passing

6

C modules bit-identical to Python

3 %

Kalman SoC error vs 7.2 %/week
naive

13 / 13

grid dips detected

PROVEN IN CODE

- int8 model exported + golden self-test in C
- autopilot: 50 / 50 C scenarios, zero chatter
- signed log: tamper, delete, replay caught

SYNTHETIC ONLY

- SoH 8.2 pt on simulated tubular batteries
- RUL band calibrated (0.99) but wide
- NILM / PQ on synthetic streams

NEEDS HARDWARE / DATA

- ESP32 flash + latency on silicon
- real bench: battery, CT, relays
- aging campaign → real SoH accuracy

Sentinel pays off for the family and for V-Guard.

One home with a ₹15,000 battery, and V-Guard across 1 lakh such homes.

THE FAMILY

2.5×

longer backup for the fridge, Wi-Fi and lights

6–14 wk

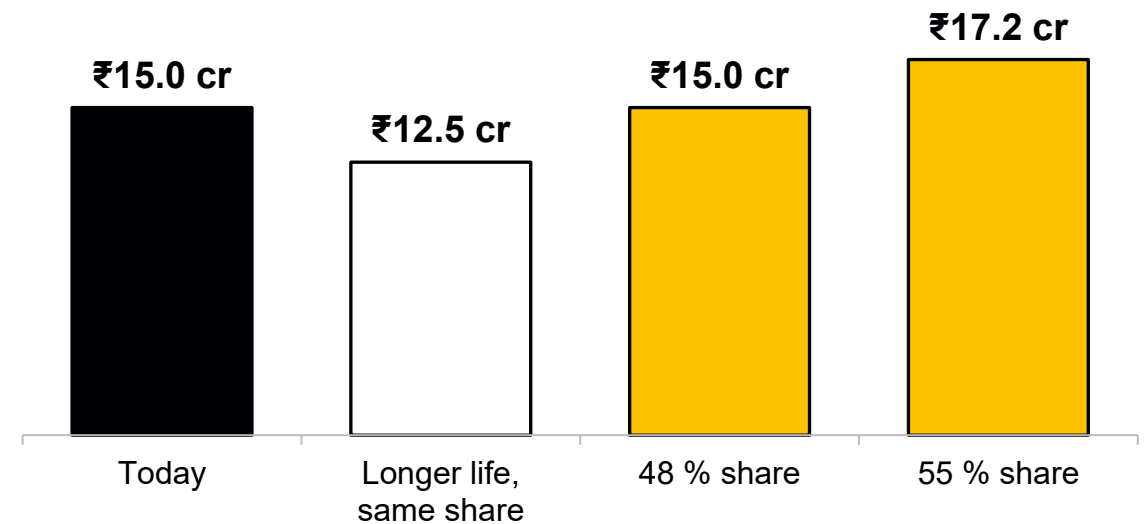
warning before the battery dies

₹6,250

less spent on batteries per decade*

V-GUARD

battery revenue from 1 lakh homes, ₹ cr a year



Break-even at 48 % share of replacements

1 % less warranty cost ≈ ₹0.7 cr a year

* Embedded, if charge control adds 20 % life (to be measured). Model: 4-year battery life, 40 % of replacements to V-Guard today (assumed). Warranty: V-Guard Annual Report FY24.

A growing market, and fewer batteries in scrap yards.

India's home backup still runs on lead-acid.

US\$ 487 M

home-UPS market
by 2030

~53 %

of inverter batteries
are lead-acid

60–80 %

of used lead-acid is
recycled informally

~42,000

fewer batteries per lakh
homes, per decade*

1

Planned replacements

The warning window becomes a
planned sale.

2

Take-back at the warning

Old batteries go to authorised
recyclers.

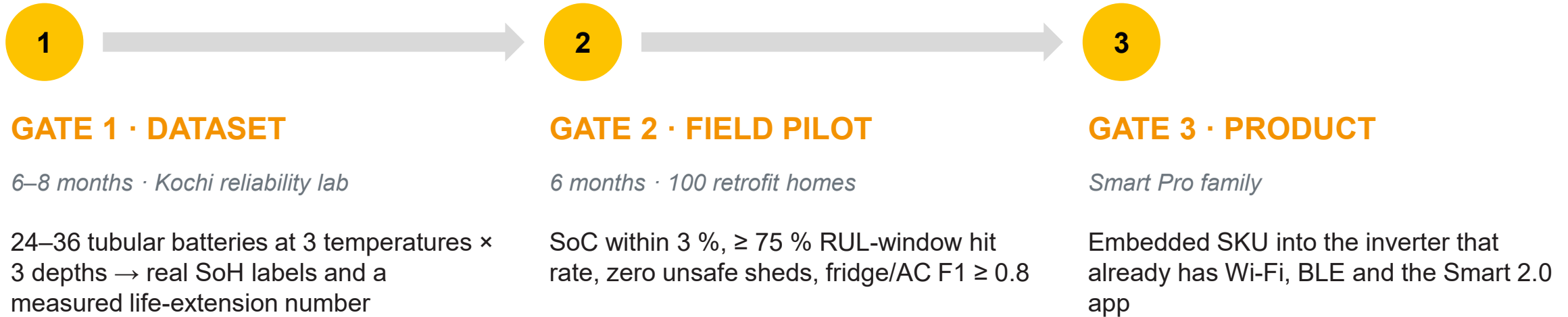
3

Upgrade, don't replace

Retrofit makes today's inverters
intelligent.

* 1 lakh homes, 4-year life today, +20 % with Sentinel-Embedded. Sources: MarketsandMarkets (home UPS), Mordor Intelligence (chemistry share), industry recycling estimates.

Three validation gates — and what we need to pass the first.



What V-Guard gains

battery replacements at the right time · warranty by evidence · a premium 'intelligent' tier · a fleet-data moat

What we ask for

one opened Prime-series inverter + charger schematic · ESP32 bench parts · batteries and cyclers for Gate 1



Isolation

reinforced AMC1311 barrier; HV zone behind a moat; TVS on every line

Fail-safe

NC contactors — any fault, reset or brown-out brings loads back

Standards

IS 13252 / IEC 62368-1 (BIS gate) · IEC 62040 UPS · IEC 61000 EMC · IEEE 1159 / IEC 61000-4-30 Class-S-like PQ · UL94 V-0 enclosure

Charging

-24 mV/°C compensation, derate above 45 °C, stop at 58 °C, 2 s heartbeat revert to factory

**Coil off means loads on.
A window, never a date.
Offline first, fleet second.**

Thank you — we'd love your questions.
Team Codey Tingle (TI3271)

